

# **Han Le: MC311 Requests Pattern Analysis Final Report**

## **Introduction and Project Overview**

People call 911 when they need assistance in urgent crises, but most people have never heard of 311. 311 allow residents to call in non-emergency service requests that affect their quality of life, such as road services, housing, waste management, snow removal, etc. Because these requests reflect both public needs and county response operations, my analysis will provide valuable insight into service efficiency and patterns.

The primary dataset I used was the MC311 Requests dataset from DataMontgomery. Additional datasets I used was Maryland daily weather data from the National Oceanic and Atmospheric Administration (NOAA), and ZIP Code Tabulation Area, which had demographic information based on cities, from Maryland.gov. I also had to merge multiple datasets on income data from Data Commons to get one full dataset on just income.

The goal of this project was to analyze MC311 service requests and explore how factors such as weather, location, demographics, and time influence service and response outcomes. Python was the tool I used for the entire project on Google Colab. I used libraries such as Pandas, NumPy, Matplotlib, and Scikit-learn for data wrangling, visualization, statistical analysis, and predictive modeling

## **Data Ingestion**

The MC311 dataset was filtered to include requests submitted from January 2025 through April 2026, as the original dataset has over 7.7 million. To make the analysis more focused and manageable, I filtered the dataset to specifically focus on five cities (Brookeville, Burtonsville, Colesville, Eastern Silver Spring, and Rockville) and three service areas (Highway Services, Treasury, and Code Enforcement).

I created several new variables, one being a `response\_time` variable to measure how long requests took to resolve. The dataset did have its own response time variable but when I looked at each row, it didn't match up the days between date opened and closed so I created my own variable. Additional variables such as `day\_of\_week`, `month`, and `season` were also created. From the weather dataset, the variable `avg\_temp` was created since the original variable were all N/As.

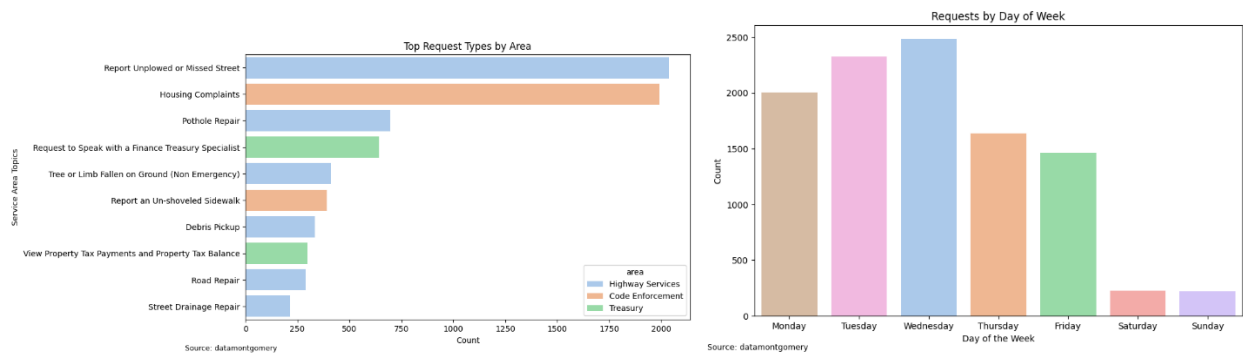
I merged all the datasets by either ZIP code or date depending on the data source. Demographic variables such as population, income, housing units, ethnicity, and median age were integrated into the final dataset. The final dataset contained over 10,000 service requests along with weather, demographic, and SLA-related information. The cleaned dataset provided a strong foundation for both exploratory analysis and predictive modeling.

Full steps and details are available in my project GitHub repository:  
<https://github.com/lnghan/MC311-Analysis>

# Han Le: MC311 Requests Pattern Analysis Final Report

## Basic Descriptive Statistics

The final dataset variables included service areas and their corresponding request topics, response times, SLA status, daily weather conditions, population statistics, income levels, and demographic information. My initial exploratory data analysis (EDA) revealed many trends. For one, Highway Services represented the largest category of service requests across nearly all cities, but request volume also varied by season and day of the week. For example, mid-week periods tended to experience higher request counts, while weekends showed extremely lower activity levels.



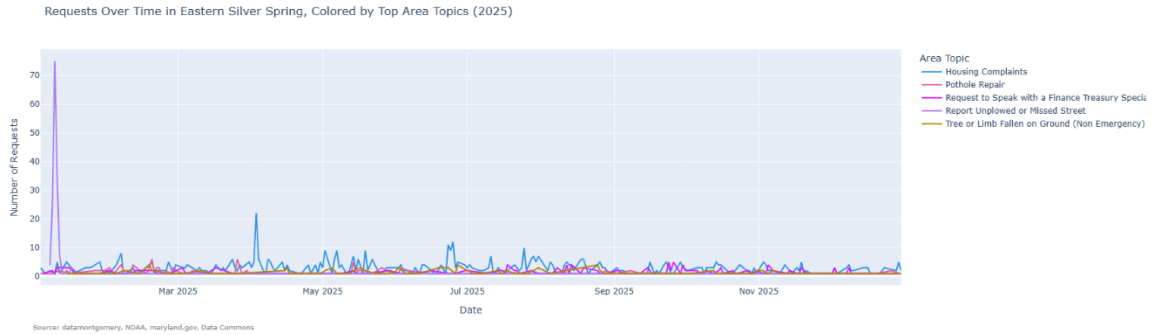
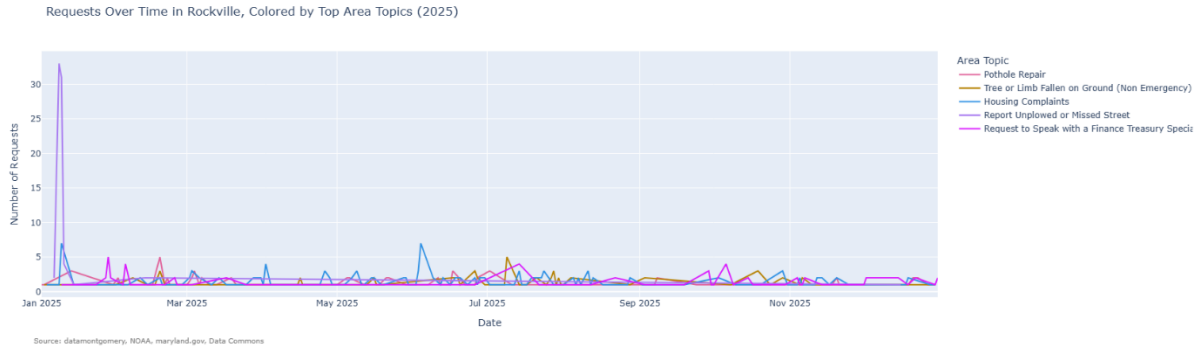
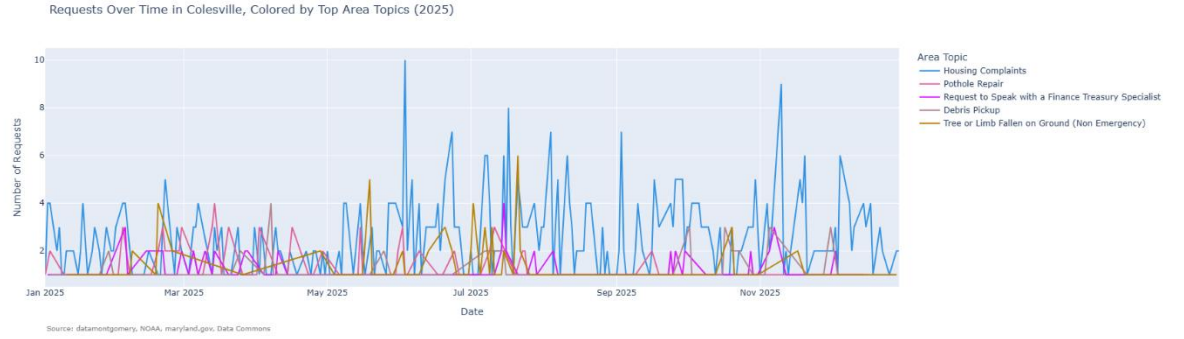
## Final Data Product

I calculated requests per capita to better understand if service demand relates to population size. Rockville showed the highest request rate per capita, meaning that residents in the area may report issues more frequently, or it experiences more service-related concerns.

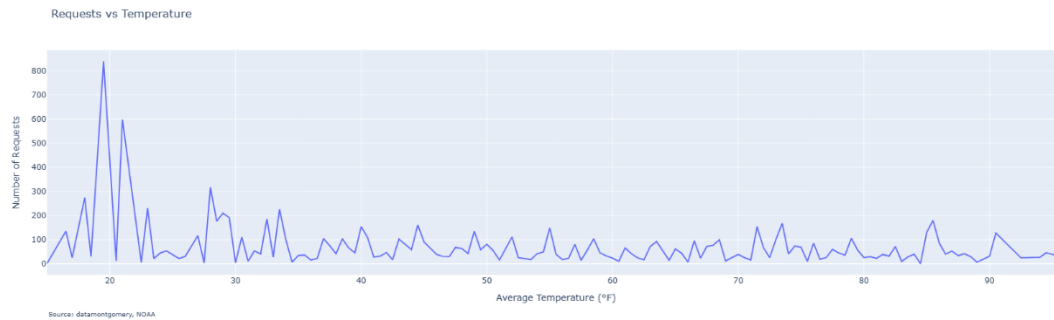


The bar chart also revealed that Highway Services consistently dominated the request categories, but certain cities experienced unique concerns. For example, Eastern Silver Spring experienced a major peak in unplowed or missed street requests during winter weather events, while Rockville showed smaller spikes in pothole repair and street drainage requests. Colesville also displayed occasional increases in requests related to fallen trees and blocked roads.

# Han Le: MC311 Requests Pattern Analysis Final Report

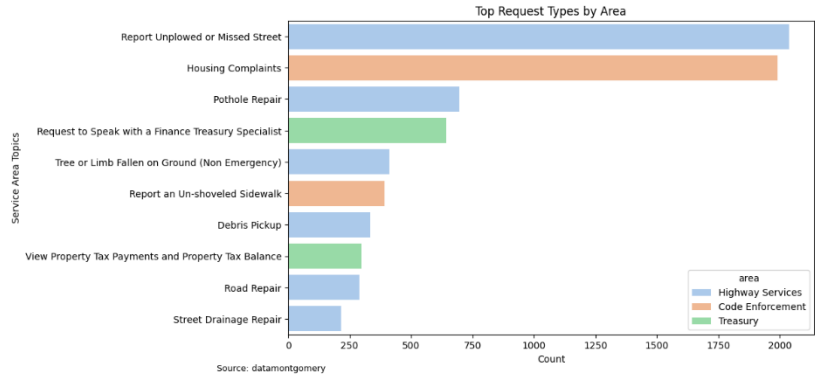


I also delved into seasonal analysis, which revealed a spike in requests during winter months, particularly in January. I could infer that the spike of MC311 calls was because of the snowfall in January. I also created a line graph mapping request volume with average temperature, which showed that colder temperatures were associated with more requests.

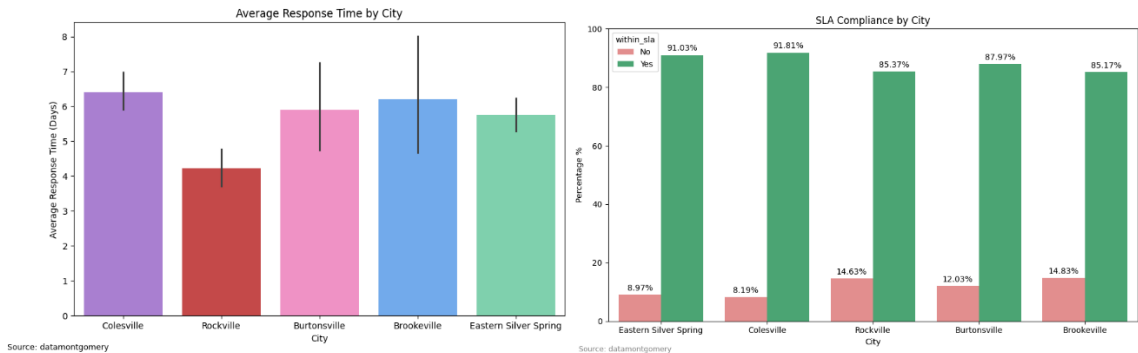


**Han Le: MC311 Requests Pattern Analysis Final Report**

To correlate with my inference, snow-related complaints, especially unplowed or missed streets, contributed heavily to winter spikes. January 2025 experienced especially high levels of winter weather service requests.



I also focused on analyzing response times and Service Level Agreement (SLA) compliance across cities. Rockville not only had a high request rate per capita but also maintained relatively quick response times compared to other cities.



To test whether SLA compliance differs by city, I performed chi-squared test, which showed a statistically significant association between city and SLA compliance, with a p-value below 0.05. Meaning location may influence whether service requests meet SLA requirements.

```
Chi-squared statistic: 82.68
P-value: 4.7143856982950015e-17
Degrees of freedom: 4

There is a statistically significant association between city and SLA compliance (reject H0).
```

Additionally, a multiple regression model using variables such as SLA days, population, temperature, income, housing units, ethnicity, and age was used as indicators for response time. However, the regression model produced a R-squared value of 0.126, meaning the variables explained only a small portion of the variation in response outcomes.

A classification model, specifically a logistic regression model, was created to predict whether requests would meet SLA requirements. The model achieved an overall accuracy of 97%, but this may be because the dataset was highly imbalanced, as most requests were classified as successful SLA outcomes. The F1-score for successful SLA outcomes was 0.98, while the F1-score for unsuccessful SLA outcomes was lower at 0.81. Additionally, the recall score for

|              | precision | recall | f1-score | support |
|--------------|-----------|--------|----------|---------|
| 0            | 0.88      | 0.76   | 0.81     | 200     |
| 1            | 0.97      | 0.99   | 0.98     | 1829    |
| accuracy     |           |        | 0.97     | 2029    |
| macro avg    | 0.93      | 0.87   | 0.90     | 2029    |
| weighted avg | 0.96      | 0.97   | 0.96     | 2029    |

## **Han Le: MC311 Requests Pattern Analysis Final Report**

unsuccessful SLA cases was only 0.76, meaning the model missed some requests that failed to meet SLA requirements.

### **Reflection**

What I felt was rewarding throughout the project was being able to successfully integrate multiple datasets together using dates and ZIP codes. Combining public service data with weather and demographic information made it possible for me uncover patterns that would not have been visible using a single dataset alone. I was also glad to discover meaningful trends; the relationship between weather and service demand was especially interesting because it showed how environmental conditions can directly impact infrastructure and operations.

I encountered several challenges, one of them being missing data, particularly with the income datasets. My classification model also suffered from class imbalance, which affected its ability to accurately identify unsuccessful SLA cases. I was also disappointed that my regression model had relatively weak predictive power, suggesting that additional variables may be needed to better explain response outcomes.

In the future, I would use more advanced machine learning techniques and resampling methods to address the class imbalance and improve classification accuracy.

### **Next Steps and Recommendations**

For my next steps, I want to include additional ZIP codes and a longer time range. Finding datasets that has operational variables related to MC311, such as request difficulty, staffing, etc, could also improve my analysis and strengthen my predictive models.

My findings may help Montgomery County improve resource allocation during peak seasons, particularly during winter weather events, as it was shown that that's when service requests increase significantly. Geographic differences in request patterns may also help county agencies better tailor services to the needs of specific communities.

### **Acknowledgements**

Special thanks to Professor Alraee, Professor Perine, Professor Park, the Montgomery County Government, the DataMontgomery team, DATA205 classmates, and the Data Science Early College Cohort for their support throughout this project!

## **Han Le: MC311 Requests Pattern Analysis Final Report**

### **References**

Centers, National. “Daily Summaries Station Details: MARYLAND SCIENCE CENTER, MD US, GHCND:USW00093784 | Climate Data Online (CDO) | National Climatic Data Center (NCDC).” Noaa.gov, Climate Data Online (CDO), 2026, [www.ncdc.noaa.gov/cdo-web/datasets/GHCND/stations/GHCND:USW00093784/detail](http://www.ncdc.noaa.gov/cdo-web/datasets/GHCND/stations/GHCND:USW00093784/detail).

Data Commons. *Datacommons.org*, [datacommons.org/ranking/Median\\_Income\\_Household/CensusZipCodeTabulationArea/geoId/24?h=zip/21640&unit=](https://datacommons.org/ranking/Median_Income_Household/CensusZipCodeTabulationArea/geoId/24?h=zip/21640&unit=). Accessed 9 May 2026.

“Maryland Census Data - ZIP Code Tabulation Areas (ZCTAs).” *Maryland.gov*, 2020, [data.imap.maryland.gov/datasets/maryland::maryland-census-data-zip-code-tabulation-areas-zctas/explore?location=38.823450%2C-77.268400%2C7&showTable=true](https://data.imap.maryland.gov/datasets/maryland::maryland-census-data-zip-code-tabulation-areas-zctas/explore?location=38.823450%2C-77.268400%2C7&showTable=true). Accessed 9 May 2026.

*Montgomerycountymd.gov*, [data.montgomerycountymd.gov/Government/MC311-Service-Requests/xyth-brr2/about\\_data](https://data.montgomerycountymd.gov/Government/MC311-Service-Requests/xyth-brr2/about_data)